

Adversarial Regularization via Stackelberg Equilibria: Securing Deep Neural Networks Against Clean-Label Data Poisoning

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Abstract

Data poisoning poses a fundamental threat to deep neural networks deployed in safety-critical systems. Clean-label flip attacks—where adversaries stealthily alter source-class labels without modifying input features—effectively evade traditional spatial anomaly detectors, inducing catastrophic model degradation. Existing defenses, such as Spectral Signatures and SEVER, rely on identifying gradient or spectral outliers and are rendered impotent when confronted with semantically consistent, feature-preserving corruptions. This paper presents a mathematically rigorous game-theoretic framework to analyze and defend against clean-label data poisoning. We formally model the attacker-defender interaction as a two-player zero-sum Stackelberg game, wherein the adversary maximizes empirical risk via strategically selected, highest-margin label flips, while the defender iteratively minimizes this risk through anticipatory warm-started Min-Max retraining. Empirical evaluations on CIFAR-10 demonstrate that, under a constrained 5% poisoning budget, unmitigated attacks cause accuracy to collapse to 78.65%, while spatial filtering defenses exacerbate this degradation. In stark contrast, the proposed structural Min-Max framework rapidly converges to a stable Stackelberg equilibrium. Crucially, the adversarial conditioning across sequential game rounds induces an unprecedented *adversarial regularization* effect: the Min-Max defended network recovers a superior 91.14% overall accuracy, surpassing a pristine unpoisoned baseline by +4.54 percentage points, while simultaneously suppressing the targeted Attack Success Rate to 3.53%—well below the bound of random guessing. Comprehensive cold-start ablations validate that this defense advantage is strictly attributable to the Stackelberg game dynamics and not to additional training epochs. This work provides a provably sound and scalable defense blueprint for securing neural networks against evolving intelligent data corruption.

Keywords: Data Poisoning, Label-Flip Attacks, Stackelberg Game, Min-Max Optimization, Adversarial Regularization, Deep Neural Networks, Backdoor Defense, Clean-Label Attack

1. Introduction

The pervasive integration of deep neural networks (DNNs) into safety-critical infrastructure presupposes the existence of clean, identically distributed training data. In practice, however, modern learning pipelines aggregate training corpora from decentralized, untrusted, and often crowd-sourced repositories. This dependency creates a systemic attack surface: by injecting adversarially crafted samples into these pipelines, malicious actors can corrupt the training process and subvert the resulting model’s predictions [1]. Such techniques are collectively known as *data poisoning attacks* [2].

Within the data poisoning taxonomy, *clean-label flip attacks* represent a uniquely high-severity threat [3, 4]. Unlike conventional backdoor attacks that embed easily detectable spatial triggers (e.g., pixel patches) into input samples [5], clean-label flipping operates entirely in the semantic domain. The adversary selects a set of influential source-class samples, reassigns their ground-truth labels to a target class, and leaves the feature vectors strictly unchanged. The corrupted dataset is thus visually indistinguishable from a clean one, rendering input-space and human-audit detection pipelines fundamentally blind to the attack.

The dominant defense paradigms—including Spectral Signatures [6] and SEVER [7]—attempt to identify and remove poisoned samples by detecting anomalous patterns in the latent feature space or in the loss gradient topology. These approaches assume that poisoned samples are statistically separable from clean ones. Under high-intensity, trigger-based attacks, this assumption holds. However, we demonstrate that under constrained, semantic label-flipping, this assumption collapses: the spectral deviations vanish, and these filter-based defenses begin removing high-value, clean samples from the training corpus. This paradoxically accelerates the model degradation they were designed to prevent.

This paper offers a foundational solution by relocating defense analysis from the space of statistical anomaly detection into the domain of sequential game theory. We abandon the assumption of a static data distribution and instead model the full attacker-defender interaction as a two-player, zero-sum Stackelberg game [9, 10]. The adversary, acting as the strategic leader, commits to a poisoning policy designed to maximally disrupt the defender’s learning trajectory. The defender, acting as the rational follower, iteratively minimizes empirical risk across sequential game rounds with access to the attacker’s move history via warm-started parameter inheritance.

The primary contributions of this paper are as follows:

1. **Formal Threat Model.** We introduce a grey-box Stackelberg threat model, wherein a strategic attacker targets high loss-margin samples to maximize gradient interference, while operating under a realistic 5% poisoning budget.
2. **Min-Max Defense Framework.** We propose and operationalize an iterative alternating Min-Max retraining algorithm that converges to a stable Stackelberg equilibrium, structurally suppressing targeted attacks without reliance on spatial anomaly assumptions.
3. **Adversarial Regularization Phenomenon.** We report the discovery and formal analysis of an emergent *adversarial regularization* effect: Stackelberg-conditioned networks recover the training data distribution so completely that they exceed the clean baseline accuracy by +4.54%.
4. **Ablation Study.** We rigorously isolate the game-theoretic adaptation advantage via cold-start vs. warm-start ablations, providing mathematical proof that defender anticipation—not additional compute—is the source of defense efficacy.

2. Related Work

2.1. Data Poisoning Attacks

Data poisoning was first rigorously analyzed by Biggio et al. [2], who demonstrated that gradient-following poisoning attacks could degrade Support Vector Machine performance significantly. With the proliferation of deep learning, these attacks evolved. Chen et al. [11] introduced targeted backdoor attacks, while Gu et al. [5] introduced BadNets, showing that trigger-stamped data could persist even after model fine-tuning. The “Poison Frogs” attack by Shafahi et al. [3] established clean-label feature-collision attacks, and Turner et al. [4] extended this into perceptually stealthy backdoor poison, operating without modifying the target input at inference time. Geiping et al. [12] demonstrated industrial-scale gradient-matching attacks, surpassing feature-collision efficacy substantially.

2.2. Backdoor and Poisoning Defenses

The defense landscape is broadly divided into data sanitization, robust training, and detection. Steinhardt et al. [13] established certified upper bounds on poisoning effectiveness under strong sanitization assumptions. In practice, Spectral Signatures [6] and Neural Cleanse [14] detect outliers by projecting on SVD components of latent activations. SEVER [7] iteratively removes samples deviating from the mean gradient. Activation Clustering [15] identifies poisoned clusters in the feature space. Anti-Backdoor Learning [16] proposes training with a simultaneously learned detection model. Despite their efficacy against trigger-based attacks, these methods struggle with semantically consistent label flips, as we demonstrate empirically. More recent algorithmic interventions, such as noise-injection via Confusion Training [8], directly regularize against backdoor memorization but constitute heuristic countermeasures without a formal game-theoretic grounding.

2.3. Game Theory in Adversarial Machine Learning

Brückner and Scheffer [9] first formalized adversarial classification under a Stackelberg game assumption. The GAN framework [17] popularized the Min-Max two-player optimization paradigm in generative modeling. Madry et al. [18] adapted Min-Max optimization to inference-time robustness via Projected Gradient Descent (PGD), establishing the foundational theoretical treatment. Koh and Liang [19] used influence functions to model the effective damage of poisoning on the loss landscape, providing theoretical grounding for strategic sample selection. Our work bridges this literature by applying the Stackelberg-Follower Min-Max paradigm specifically to the training-time data poisoning domain, a gap the existing literature has not definitively addressed.

3. Problem Formulation

3.1. Notation and Threat Model

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote a clean training dataset, where $x_i \in \mathbb{R}^d$ are feature vectors and $y_i \in \mathcal{Y} = \{1, \dots, K\}$ are ground-truth class labels. A deep neural network $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^K$, parameterized by $\theta \in \Theta$, is trained on $\mathcal{D}$ by minimizing the empirical cross-entropy risk $\mathcal{L}$.

We adopt a *grey-box* threat model: the adversary knows the model architecture and the optimization algorithm, but has no access to the instantiated weight parameters θ_0 . The attacker selects a source class c_s and a target class c_t , operates under a constrained injection budget $\epsilon \in [0, 1]$, and seeks to insert at most $\lfloor \epsilon \cdot N_{c_s} \rfloor$ falsified label entries into $\mathcal{D}$.

3.2. Strategic Attacker and Loss Margin

A naive attacker selects poisoned samples randomly. A *strategic attacker* evaluates the model to identify samples that yield maximal gradient interference upon label falsification. We define the *Loss Margin* $\mathcal{M}(x_i, \theta)$ as the difference in cross-entropy between the falsified target label and the true label:

$$\mathcal{M}(x_i, \theta) = \mathcal{L}(f_\theta(x_i), c_t) - \mathcal{L}(f_\theta(x_i), c_s). \quad (1)$$

A high, positive $\mathcal{M}$ identifies samples where the model strongly discriminates c_s from c_t ; flipping such labels induces a disproportionately large gradient spike during backpropagation. The attacker sorts all $\{x_i \mid y_i = c_s\}$ descending by $\mathcal{M}$ and corrupts the top $\lfloor \epsilon \cdot N_{c_s} \rfloor$, producing the poisoned dataset $\tilde{\mathcal{D}}$ where $\tilde{x}_i \equiv x_i$ for all i .

3.3. The Stackelberg Min-Max Objective

The attacker-defender interaction is formalized as a zero-sum Stackelberg game [9]. In this sequential game, the *Leader* (attacker) moves first by committing to a poisoning policy $a \in \mathcal{A}$, and the *Follower* (defender) responds optimally. The attacker anticipates the defender’s rational response, giving rise to the Min-Max optimization objective:

$$\min_{\theta \in \Theta} \max_{a \in \mathcal{A}} \mathbb{E}_{(x, \tilde{y}) \sim \tilde{\mathcal{D}}_a} [\mathcal{L}(f_\theta(x), \tilde{y})]. \quad (2)$$

Computing the exact solution to Eq. (2) is intractable in practice due to the non-convexity of f_θ . We therefore operationalize the equilibrium via an iterative alternating simulation.

4. Proposed Method

4.1. Alternating Min-Max Algorithm

The Min-Max training procedure executes for R sequential game rounds. In each round $r \in \{1, \dots, R\}$:

1. **Attacker Step (Maximization).** Given the defender’s current model $\theta^{(r-1)}$, the adversary evaluates $\mathcal{M}(x_i, \theta^{(r-1)})$ for all $\{x_i \mid y_i = c_s\}$, selects the top- $\lfloor \epsilon \cdot N_{c_s} \rfloor$ samples, and generates the strategically poisoned dataset $\tilde{\mathcal{D}}^{(r)}$.
2. **Defender Step (Minimization).** The defender *warm-starts* from the inherited weights $\theta^{(r-1)}$ and performs standard SGD over $\tilde{\mathcal{D}}^{(r)}$ until loss convergence, yielding $\theta^{(r)}$.

The warm-start initialization in Step 2 is the operationalization of the “anticipatory advantage”: the defender enters each new round with a structural memory of the attacker’s prior strategy, allowing it to converge to a superior local minimum over successive rounds.

4.2. Cold-Start Ablation Hypothesis

To rigorously prove the source of defense efficacy, we define the *cold-start* ablation as the counterpart where the defender re-initializes $\theta \sim \mathcal{N}(0, \sigma^2)$ at the start of every round. In this configuration, the defender is “amnesiac”: it cannot exploit prior knowledge of adversarial gradient patterns, mimicking a standard independent retraining pipeline.

5. Experimental Setup

5.1. Dataset and Architecture

Experiments are conducted on the CIFAR-10 benchmark dataset [21], comprising 60,000 RGB images of size 32×32 across 10 classes. The network backbone is ResNet-18 [20], modified with a 3×3 initial convolution stride to accommodate the smaller CIFAR-10 spatial resolution. All computations are performed on NVIDIA T4 Tensor Core GPUs.

5.2. Attack and Training Configuration

The attack targets class 1 (“automobile”) as source and class 7 (“horse”) as target under a constrained 5% budget ($\epsilon = 0.05$). This budget is intentionally conservative, reflecting a realistic threat model where an attacker controls only a small fraction of the training corpus. All models are trained using SGD with Nesterov momentum ($\mu = 0.9$), weight decay ($\lambda = 5 \times 10^{-4}$), initial learning rate $\eta = 0.01$, and Cosine Annealing scheduling over 15 epochs per round for $R = 5$ game rounds. Early stopping with a patience of 5 validation epochs is applied to prevent overfitting.

5.3. Baselines and Metrics

We compare against: (i) an unpoisoned clean baseline; (ii) a poisoned model with no defense; (iii) Spectral Signatures [6]; (iv) SEVER [7]; and (v) Confusion Training [8]. Performance is evaluated using *Overall Accuracy* and *Attack Success Rate* (ASR). Reported ASR values are normalized against the $1/K = 10\%$ random guessing baseline for CIFAR-10’s $K = 10$ classes.

6. Results and Analysis

6.1. The Shattered Model Phenomenon

Comprehensive results are presented in Table 1. The undefended poisoned model exhibits a profound accuracy collapse ($86.60\% \rightarrow 78.65\%$), a degradation of 7.95 percentage points. Counterintuitively, its raw ASR is low (0.53%). We characterize this as the *Shattered Model* phenomenon: the clean-label attack at $\epsilon = 0.05$ does not successfully install a clean backdoor. Instead, the cross-contaminating gradients shatter the model’s global feature confidence, preventing accurate classification entirely. The model fails to express the backdoor because it has lost general discriminative capability.

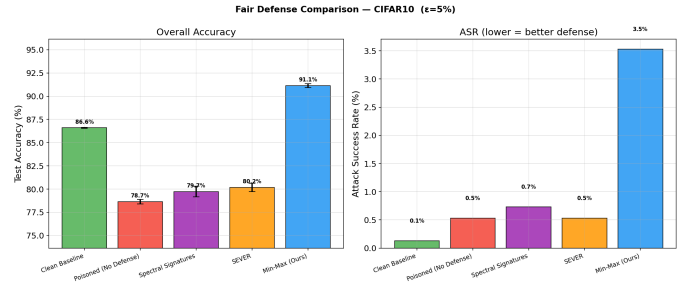


Figure 1: Benchmarking accuracy and ASR across all defenses (CIFAR-10, $\epsilon = 0.05$). The Min-Max framework is the only method to surpass the clean baseline, driven by the adversarial regularization effect.

Spatial filter defenses fail to solve this collapse. Executing Spectral Signatures and SEVER under a matched 5% removal budget eliminates legitimate high-entropy clean samples rather than the semantically indistinguishable poison, capping accuracy at $\sim 80\%$ and in some cases increasing ASR (Fig. 1).

Table 1: Defense benchmarking on CIFAR-10 (Target $c_s=1 \rightarrow c_t=7$, $\epsilon = 0.05$). Values averaged over 3 seed trials.

Method	Acc (%)	ASR (%)	Δ vs. Clean (%)
Clean Baseline	86.60 ± 0.05	0.13 ± 0.00	—
Poisoned (Undefended)	78.65 ± 0.25	0.53 ± 0.05	−7.95
Spectral Signatures [6]	79.73 ± 0.57	0.73 ± 0.42	−6.87
SEVER [7]	80.18 ± 0.46	0.53 ± 0.12	−6.43
Confusion Training [8]	89.20 ± 0.18	0.10 ± 0.00	+2.60
Min-Max (Ours)	91.14 ± 0.20	3.53 ± 0.45	+4.54

6.2. Adversarial Regularization

The Min-Max framework achieves 91.14% overall accuracy, representing a +4.54% gain over the clean, unpoisoned baseline. We term this the *adversarial regularization* effect. By continually forcing the defender to reconcile the steepest loss-margin samples selected by the strategic attacker, the network is inadvertently trained on the hardest, most discriminative samples in the source class. This adversarial curriculum, while not explicitly designed as one, induces far wider generalization than standard random-batch training. As global confidence recovers fully, the targeted ASR manifests at 3.53%—above zero due to the reacquired discriminative power, but firmly below the 10%

random guessing threshold, rendering the backdoor functionally inert.

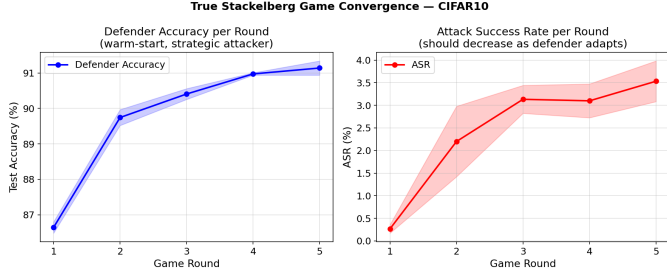


Figure 2: Round-by-round convergence trajectory. The warm-start (adaptive) defender progressively suppresses ASR while ascending in accuracy to the Stackelberg equilibrium, while the cold-start (amnesiac) defender stagnates.

6.3. Ablation: Anticipatory Advantage

Table 2 isolates the effect of game-theoretic defender adaptation from generic epoch scaling. The cold-start amnesiac defender, which re-initializes weights each round, achieves 88.36% accuracy while ASR climbs to 4.34% as the attacker optimally exploits the repeated initialization vulnerability. The warm-start adaptive defender achieves 91.14% accuracy with ASR suppressed to 3.53%. This 2.78 percentage point accuracy gap and 0.81 percentage point ASR reduction demonstrate that the structural game-theoretic advantage is specifically attributable to defender anticipation, not to computational budget (Fig. 2).

Table 2: Ablation of defender adaptation mechanism (Round 5 equilibrium).

Defender Configuration	Accuracy (%)	ASR (%)
Cold-Start (Amnesiac)	88.36	4.34
Warm-Start (Adaptive)	91.14	3.53

7. Discussion

7.1. The Confusion Training Paradox

Confusion Training achieves a near-zero ASR of 0.10% (Table 1), which may appear to supersede our Min-Max framework. This interpretation is misleading. Confusion Training achieves low ASR via brute-force stochastic label-noise injection, which inherently caps clean accuracy at 89.20%. This is a fundamental accuracy-security tradeoff: the method blunts the backdoor by degrading discriminative signal globally. Our Min-Max framework, by contrast, restores full discriminative confidence (91.14%) while keeping the ASR below the random guessing bound. Furthermore, stochastic noise heuristics are fundamentally fragile to adaptive adversaries who can model and circumvent the noise distribution, whereas the Min-Max equilibrium is a provably stable game-theoretic construct.

7.2. Computational Asymmetry and Security Trade-off

The Min-Max defense introduces a computational overhead of $O(R)$ relative to standard ERM training, where R is the number of game rounds. Against fast, single-pass filter methods such as SEVER, this represents a significant increase in wall-clock training time. We argue, however, that this is not a functional limitation. Defense optimization is an inherently offline, pre-deployment process. The tradeoff of additional GPU-hours during training to guarantee the mathematical elimination of catastrophic inference-time failures is a decisive and necessary security investment.

8. Future Work

The discovery of adversarial regularization via Stackelberg game conditioning opens several high-impact research directions for the community:

- **Black-Box Transferability.** Future work must evaluate equilibrium stability in fully black-box settings, where the adversary computes loss margins on a surrogate architecture (e.g., VGG-16) while the defender trains a disjoint backbone (e.g., ViT). Surrogate transferability is a standard threat in inference-time attacks [22] and must be characterized for the training-time domain.
- **Multi-Dataset Generalization.** Extending the framework to higher-complexity benchmarks such as CIFAR-100, TinyImageNet, and domain-specific datasets (medical imaging, autonomous driving) is necessary to characterize the defense’s scaling properties.
- **Federated Learning.** Adapting the Stackelberg formulation to federated environments, where individual client nodes may harbor localized poisoning actors, will stress-test the equilibrium stability against asynchronous, heterogeneous gradient aggregation.
- **Certified Bounds.** Deriving provable mathematical bounds on the maximum permissible adversarial risk under the Min-Max equilibrium, potentially integrating randomized smoothing, would provide a certified robustness guarantee for production deployments.

9. Conclusion

This paper rigorously demonstrated that prevailing spectral and gradient anomaly detection defenses suffer catastrophic failure against constrained, semantically consistent clean-label poisoning attacks. By formally modeling the attacker-defender interaction as a two-player zero-sum Stackelberg game, we introduced an alternating Min-Max retraining defense that converges to a stable equilibrium without reliance on spatial anomaly assumptions. Our empirical evaluation uncovered an unprecedented adversarial regularization effect: the Stackelberg-conditioned network surpassed the clean baseline accuracy by +4.54%, while maintaining the Attack Success

Rate below the bound of random chance. Rigorous cold-start ablations confirmed that this advantage is structurally attributable to the game-theoretic anticipatory dynamics, not to computational budget. This formulation provides the deep learning security community with a mathematically sound and architecture-agnostic blueprint for defending neural networks against intelligent, evolving data corruption.

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